Keshav Rajput

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Machine learning engineer, new grad. MS Computer Science, NYU Tandon, May 2026. PyTorch, LLM agents, Kaggle.
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EDUCATION

New York University, Tandon School of Engineering Sep 2024 - May 2026
Master of Science (MS) in Computer Science GPA 3.67/4.0

Indraprastha Institute of Information Technology (IIIT) Delhi 2020 - 2024
Bachelor of Technology (B.Tech) in Computer Science and Applied Mathematics, minor in Economics CGPA 8.19/10

Coursework: Machine Learning, Neuroinformatics, Foundations of Data Science, Advanced Machine Learning, Convex Optimization

PROJECTS

LFP2Vec audit | Python, PyTorch | github.com/happyc0der/lfp2vec-audit 2026
- Reproduced LFP2Vec (NeurIPS 2025) fine-tuning on public IBL and Allen Neuropixels brain recordings on a laptop: 0.74 +/- 0.07 balanced accuracy vs 0.68 published.
- Traced its cross-lab failure to a preprocessing mismatch between labs (spectra diverge 768x above 300 Hz); a 100 Hz low-pass filter restores the published margin.
- Measured calibration under lab shift (expected calibration error 0.10 in lab, 0.56 across labs, 0.35 after the fix) and added an electrode-position control; 337 tests with leakage checks in CI.

EEG dementia re-evaluation (team of 4) | Python, PyTorch, braindecode, MNE, scikit-learn 2026
github.com/happyc0der/eeg-dementia-graph-transformer
Started as a 2025 team course project; the 2026 re-evaluation, harness, models and statistics are mine.
- Re-evaluated a course project on classifying Alzheimer's, frontotemporal dementia and controls from EEG (88 subjects): subject-level splits, nested model selection, 10x5-fold CV, bootstrap CIs, permutation tests.
- Compared 16 feature pipelines with EEG foundation models (LaBraM, CBraMod, BIOT) and CNNs, pre-registered: best model 61.1% balanced accuracy vs 33% chance, and no deep model beat spectral features.
- Found a common-mode artifact holding 92% of each channel's variance and removed it with an average reference; spy-estimator leakage tests and README tables checked against result files.

Gemma 4 SWE agent (in progress) | Python, Google ADK 2026
github.com/happyc0der/gemma-swe-agent
- Built an LLM agent in Google ADK that fixes GitHub issues with Gemma 4 31B (Kaggle; since Oct 21 on a credited public bundle; final ranking pending).
- Wrote swelite, re-implementing the organizers' eval harness before its release (Docker sandboxes, ADK YAML compiler); 109 of 129 gold checks pass.
- Patched the organizers' sandbox to match the scorer's; ran the 31B via llama.cpp on Kaggle T4x2; prompt v5.8 solved 14/33 vs 13/33, within noise.

Biohub cell tracking (Finished) | Python, PyTorch, NumPy, SciPy, polars 2026
Builds on Igor Zharov's public notebook (Apache-2.0); the harness, classifiers and experiments are mine.
- Scored 0.952 on the Kaggle Biohub Cell Tracking public leaderboard on free Kaggle GPU, against 0.947 for the public notebook it builds on; finished 591st of 3,947 teams (top 15%).
- Built an offline harness that runs GPU inference once, then re-tunes post-processing locally against the official scorer on 40 held-out videos; it reproduces the notebook's score to within 0.00011.
- Trained a 3D CNN on 134 annotated cell divisions to prune false branches: predicted +0.0047 on a pre-registered holdout, measured +0.005 on the leaderboard twice.

COMPETITIONS

- **Kaggle Playground S6E9**, 2026. Out-of-fold AUC 0.9465 from nested target encodings, a LightGBM, CatBoost and MLP blend and 192 experiments judged by paired DeLong tests. Final rank 287 of 3,575 teams (top 9%), private AUC 0.94545.
- **TrafficFlowBench**, 2026. Public leaderboard 0.892 against a 0.553 baseline (LightGBM queue forecasts, PyTorch U-Nets, an NNLS origin-destination solver). In progress, ends Nov 7, 2026.
- **Money Talks 2, AI x Fintech Hackathon (Block Convey), Money Operations track**, 2026. Solo entry Ledger, a forecast agent that checks every claim against the CRM; 94% recall and 0 false alarms on 235 deals. Result pending.

SKILLS

Languages: Python, C, C++, Kotlin, TypeScript, JavaScript, SQL, Java, R, Bash, eZ80 assembly, Monkey C

ML: PyTorch, deep learning, scikit-learn, LightGBM, XGBoost, CatBoost, Hugging Face transformers, NumPy, pandas, polars

LLMs: LLM agents (Google ADK, MCP, tool calling), vLLM, Ollama, LiteLLM, LoRA/QLoRA fine-tuning (PEFT), llama.cpp/GGUF quantization

Tools: Git, CI/CD (GitHub Actions), Docker, AWS (CDK, App Runner, DynamoDB), Vercel, Linux, WSL2, Tailscale, Claude Code

LEADERSHIP AND ACTIVITIES

- Open source: ephys-mcp on PyPI and the MCP Registry (24 tools, 52 tests); six fixes to hawk-py